

Group 8 project

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1 Introduction

This investigation uses data from the 2008 - 2012 Scottish Health Survey, which collects information on the health, socio-economic, and lifestyle factors of people living in Scotland. The factors included in this dataset are: age group, sex, employment status, meets recommended vegetable intake (Yes/No), meets recommended fruit intake (Yes/No), survey year, and body mass index (BMI).

Body mass index is a quantity derived from the mass and height of a person, calculated as mass divided by the squared height, and measured in kg/m^2 . It is a useful indicator for determining if an individual is underweight (BMI under 18.5), overweight (BMI over 25), or at a healthy weight, and it is especially useful for health data.

The aim of this investigation is to determine whether the body mass index in Scotland has changed between 2008 and 2012, and to assess if there are any differences in the BMI distribution by age, gender, socio-economic factors (employment status) or lifestyle factors (meets recommended fruit and veg intake).

2 Exploratory Analysis

2.1 Data Summaries

We first inspect the balance of the data and check for missing values. The dataset contains a total of 25224 participants. There were no missing values in any of the variables used in this report, so all 25224 participants were retained for both the exploratory analysis and the subsequent modelling.

Table 1 displays the summary statistics for BMI. The mean and median are very similar (27.81 and 27.17 respectively), indicating that the distribution is approximately symmetric, while the standard deviation of 5.26 suggests a large variation in BMI across individuals. The maximum and minimum values of 12.59 and 59.11 respectively indicate a wide range of BMI values within the sample.

Table 1: Summary statistics for BMI

Mean	SD	Median	Min	Max
27.811	5.257	27.173	12.593	59.112

2.2 Distribution Of BMI

Figure 1 shows BMI's distribution is slightly right-skewed and peaks around 26-28. We can observe that the majority of values fall between 22-32, with a large portion of individuals being classifiable as overweight. Highlighted by the peak falling within the overweight grouping.

2.3 BMI and Year

Figure 2 shows the distribution of BMI across the different years of the Scottish Health Survey. The median BMI appears to remain stable over the entire period with no clear trend. The interquartile ranges and overall spread of the data are similar across the entire period. There is little evidence to suggest that BMI changed over the study period, but formal analysis is required to confirm this.

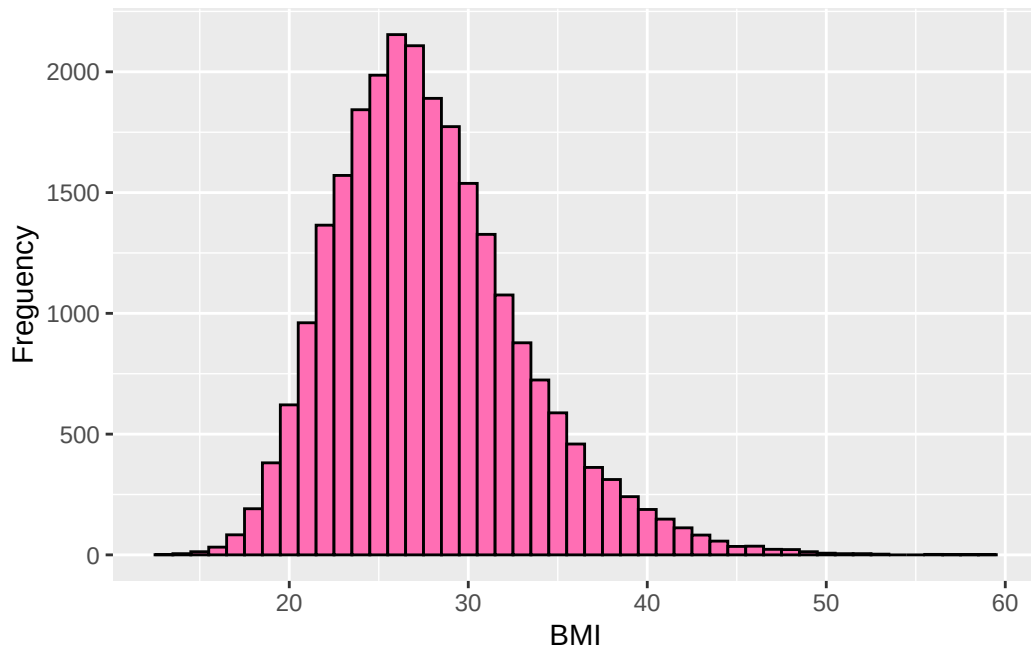


Figure 1: Histogram showing distribution of BMI.

2.4 BMI and Additional Explanatory Variables

2.4.1 Demographic Variables

Figure 3 shows the distribution of BMI by age group and sex. There appears to be a very slight increase in the median BMI as age group increases, suggesting that older individuals tend to have higher BMI values. But the interquartile ranges have a significant overlap so the differences between age groups may be small or insignificant and should further investigated in later formal analysis.

There appears to be little difference in the distribution of BMI for female and male participants, other than in the interquartile ranges, with a larger range in the female participants, indicating greater variability in BMI among females compared to males.

2.4.2 Socio-Economic and Lifestyle Variables

Figure 4 shows the distribution of BMI across the different employment categories. Individuals in full-time education appear to have a lower BMI compared to other employment groups, but the interquartile ranges for all groups substantially overlap, suggesting that the differences in BMI across these groups are relatively small and significance should be investigated with formal analysis. There also appear to be outliers at both extremes of BMI across the categories.

Figure 5 shows the distribution of BMI by fruit and vegetable consumption. The boxplots in both factors appear very similar with the medians, and interquartile ranges being at similar levels. This suggests that there is no clear evidence of a difference in BMI for participants who consume their recommended fruit and veg and those who don't, but this should be investigated in formal analysis.

3 Formal Data Analysis

We start by fitting the full multiple regression model containing all explanatory variables. The full model can be written as:

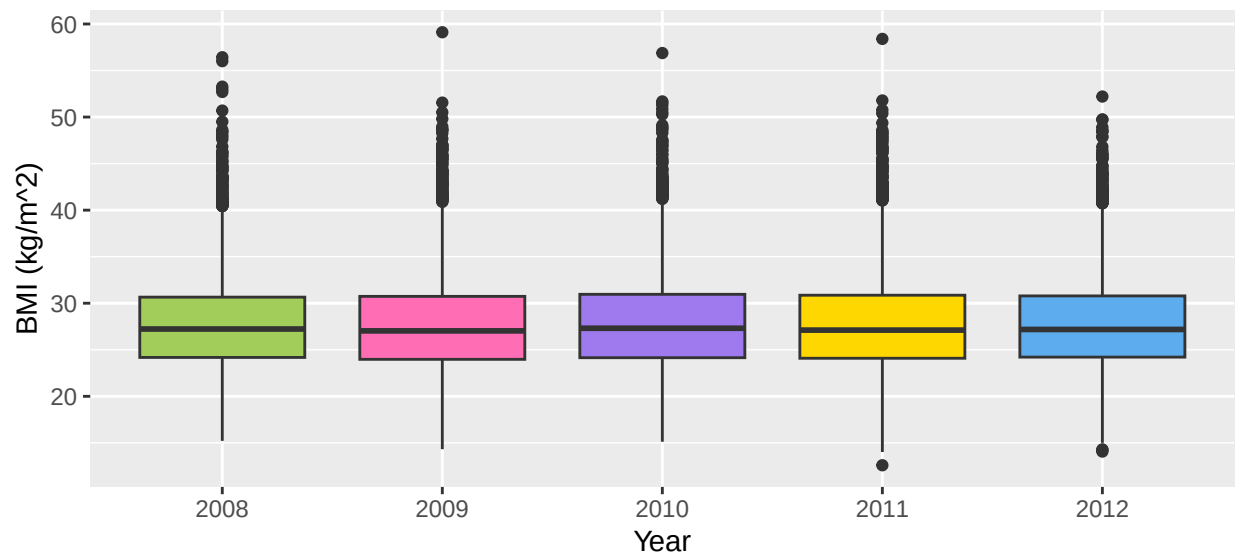


Figure 2: Boxplots of BMI for each year of survey

$$\begin{aligned}
 BMI = & \alpha + \beta_{AgeGroup} \cdot \mathbb{I}_{AgeGroup}(x) + \beta_{Sex} \cdot \mathbb{I}_{Sex}(male) \\
 & + \beta_{Employment} \cdot \mathbb{I}_{Employment}(x) + \beta_{Veg} \cdot \mathbb{I}_{Veg}(Yes) \\
 & + \beta_{Fruit} \cdot \mathbb{I}_{Fruit}(Yes) + \beta_{Year} \cdot \mathbb{I}_{Year}(x)
 \end{aligned}$$

Table 2: Estimates of the regression model coefficient

		BMI	
Predictors	Estimates		p
(Intercept)	-17.90		0.713
AgeGroup25-34	1.18		<0.001
AgeGroup35-44	2.50		<0.001
AgeGroup45-54	2.77		<0.001
AgeGroup55-64	3.28		<0.001
AgeGroup65-74	3.19		<0.001
AgeGroup [75+]	2.33		<0.001
Sex [Male]	0.05		0.411
Employment [In full-time education]	-0.87		0.016
Employment [In paid employment, self-employed or on gov't training]	0.67		0.034
Employment [Looking after home/family]	0.50		0.138
Employment [Looking for/intending to look for paid work]	0.23		0.525
Employment [Perm unable to work]	1.18		0.001
Employment [Retired]	0.66		0.045
Veg [Yes]	-0.42		<0.001
Fruit [Yes]	0.02		0.771
Year	0.02		0.377
Observations	25224		

Table 2: Estimates of the regression model coefficient

R^2 / R^2 adjusted	0.048 / 0.047
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Looking at Table 2, we can see the reduced model is:

$$\begin{aligned}
BMI = & \alpha + \beta_{Age[25-34]} \cdot \mathbb{I}_{Age}([25-34]) + \beta_{Age[35-44]} \cdot \mathbb{I}_{Age}([35-44]) \\
& + \beta_{Age[45-54]} \cdot \mathbb{I}_{Age}([45-54]) + \beta_{Age[55-64]} \cdot \mathbb{I}_{Age}([55-64]) \\
& + \beta_{Age[65-74]} \cdot \mathbb{I}_{Age}([65-74]) + \beta_{Age[75+]} \cdot \mathbb{I}_{Age}([75+]) \\
& + \beta_{Emp[FT Edu]} \cdot \mathbb{I}_{Emp}([FT Edu]) + \beta_{Emp[Employed]} \cdot \mathbb{I}_{Emp}([Employed]) \\
& + \beta_{Emp[Unable]} \cdot \mathbb{I}_{Emp}([Unable]) + \beta_{Emp[Retired]} \cdot \mathbb{I}_{Emp}([Retired]) \\
& + \beta_{Veg} \cdot \mathbb{I}_{Veg}(Yes)
\end{aligned}$$

With this reduced model, the -logistic regression model will be:

$$\begin{aligned}
BMI = & -17.9 + 1.18 \cdot \mathbb{I}_{Age}([25-34]) + 2.5 \cdot \mathbb{I}_{Age}([35-44]) + 2.77 \cdot \mathbb{I}_{Age}([45-54]) \\
& + 3.28 \cdot \mathbb{I}_{Age}([55-64]) + 3.19 \cdot \mathbb{I}_{Age}([65-74]) + 2.33 \cdot \mathbb{I}_{Age}([75+]) \\
& - 0.87 \cdot \mathbb{I}_{Emp}([FT Edu]) + 0.67 \cdot \mathbb{I}_{Emp}([Employed]) \\
& + 1.18 \cdot \mathbb{I}_{Emp}([Unable to work]) + 0.66 \cdot \mathbb{I}_{Emp}([Retired]) \\
& - 0.42 \cdot \mathbb{I}_{Veg}(Yes)
\end{aligned}$$

4 Conclusion

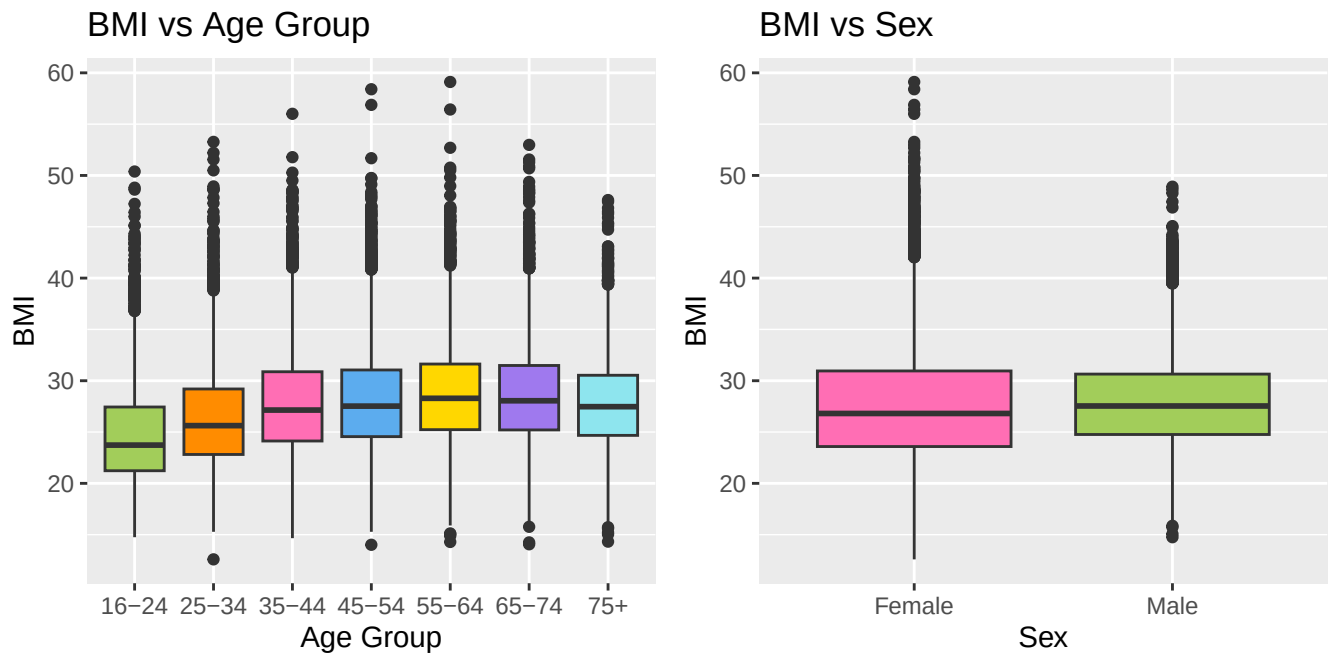


Figure 3: Boxplots of BMI across the demographic variables, age group and sex

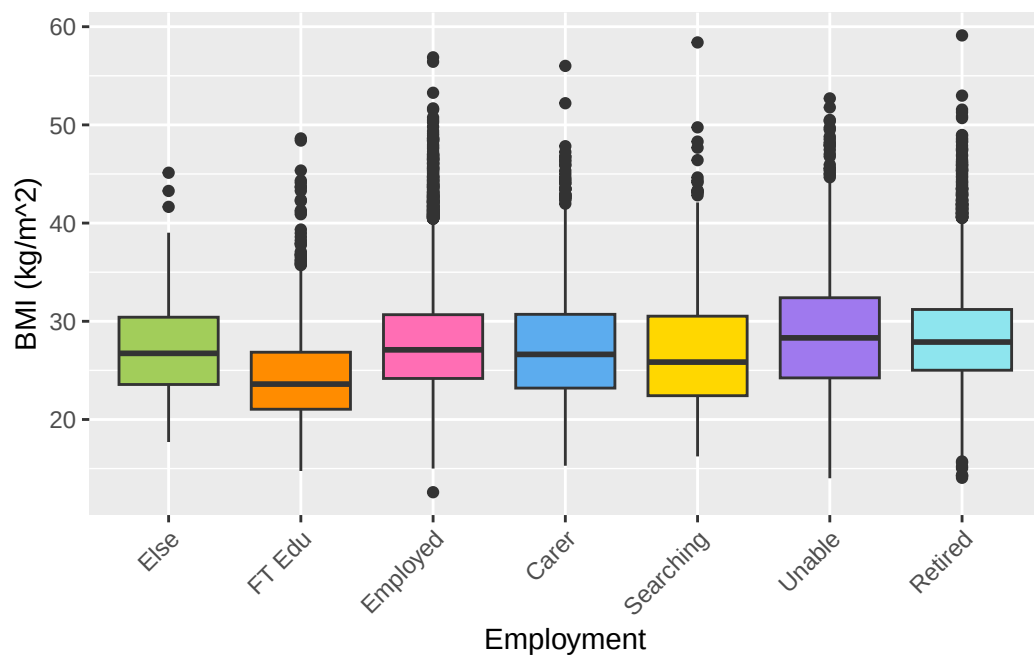


Figure 4: Boxplots of BMI across the socio-economic variable, employment

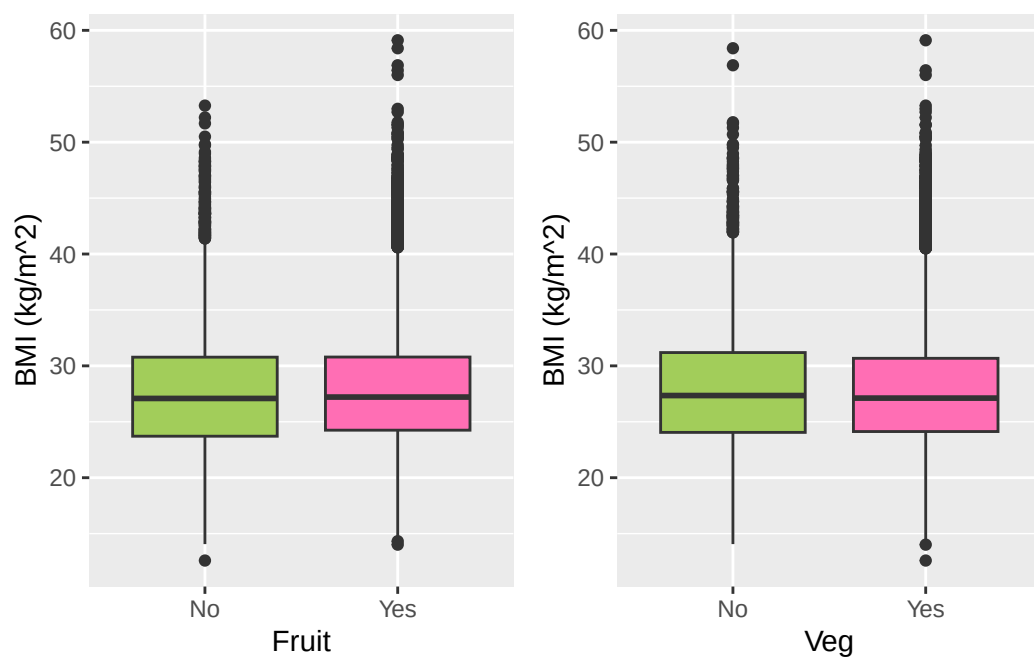
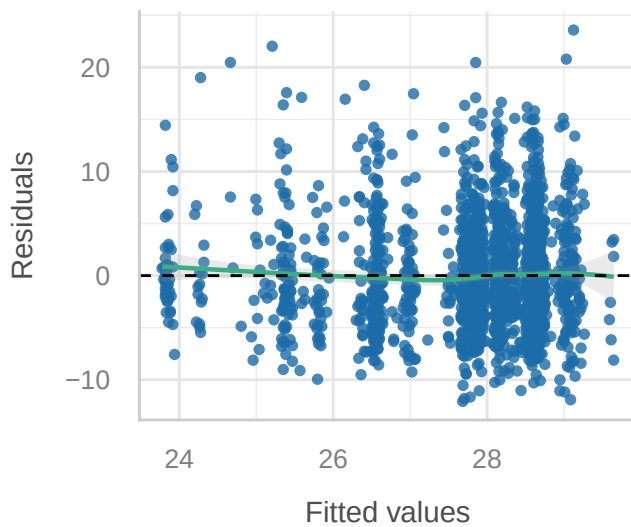


Figure 5: Boxplots of BMI across the lifestyle variables fruit and veg

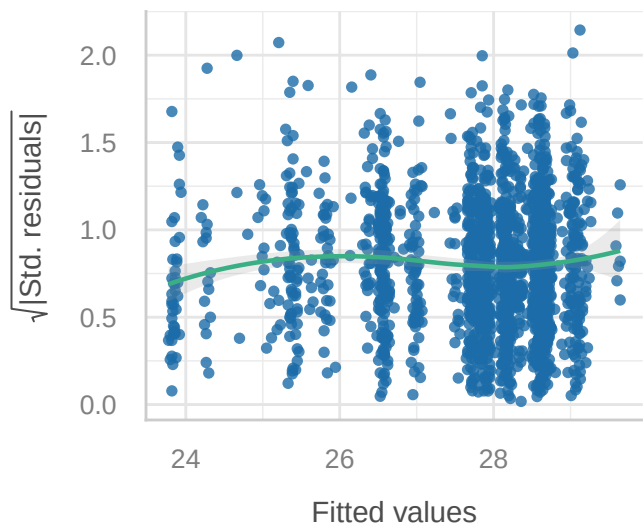
Linearity

Reference line should be flat and horizontal



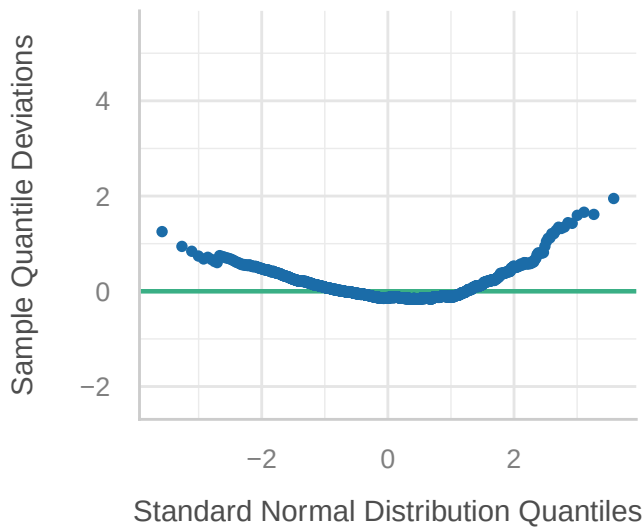
Homogeneity of Variance

Reference line should be flat and horizontal



Normality of Residuals

Dots should fall along the line



Normality of Residuals

Distribution should be close to the normal curve

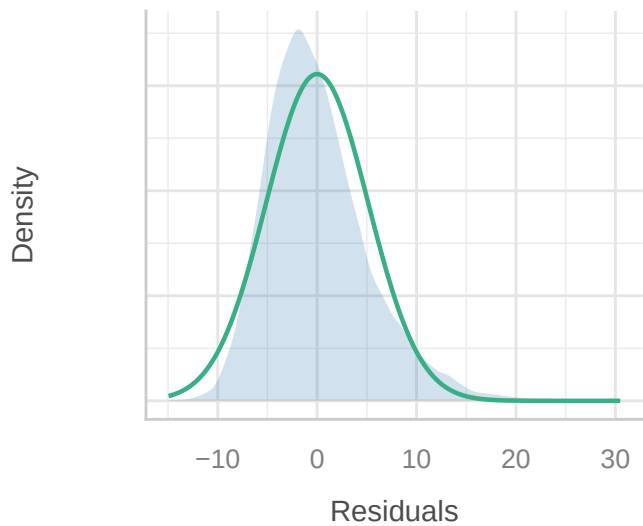


Figure 6: Final model diagnostic plots